தீர்வு

(i) $$I = \int \frac{1}{(x-2)^2+1}\,dx = \int \frac{1}{(x-2)^2+1^2}\,dx$$

$x - 2 = t$ என்க. $\Rightarrow dx = dt$

எனவே, $$I = \int \frac{1}{t^2+1^2}\,dt = \tan^{-1}(t) + c = \tan^{-1}(x-2) + c$$

(ii) $$I = \int \frac{x^2}{x^2+5}\,dx \text{ என்க.}$$

$$= \int \frac{x^2+5-5}{x^2+5}dx = \int \left(1 - \frac{5}{x^2+5}\right)dx = \int dx - \int \frac{5}{x^2+5}\,dx$$

$$= x - 5\int \frac{1}{x^2+\left(\sqrt{5}\right)^2}\,dx$$

$$= x - 5\frac{1}{\sqrt{5}}\tan^{-1}\left(\frac{x}{\sqrt{5}}\right) + c$$

$$I = x - \sqrt{5}\tan^{-1}\left(\frac{x}{\sqrt{5}}\right) + c$$

(iii) $$I = \int \frac{1}{\sqrt{1+4x^2}}\,dx = \int \frac{1}{\sqrt{1+(2x)^2}}\,dx$$

$2x = t$ என்க. $\Rightarrow 2\,dx = dt \Rightarrow dx = \frac{1}{2}dt$

எனவே, $$I = \frac{1}{2}\int \frac{1}{\sqrt{1^2+t^2}}\,dt$$

$$I = \frac{1}{2}\log\left|t + \sqrt{t^2+1}\right| + c = \frac{1}{2}\log\left|2x + \sqrt{(2x)^2+1}\right| + c$$

$$I = \frac{1}{2}\log\left|2x + \sqrt{4x^2+1}\right| + c$$

(iv) $$I = \int \frac{1}{\sqrt{4x^2-25}}\,dx = \int \frac{1}{\sqrt{(2x)^2-25}}\,dx$$

$2x = t$ என்க. $\Rightarrow 2\,dx = dt \Rightarrow dx = \frac{1}{2}dt$

எனவே, $$I = \frac{1}{2}\int \frac{1}{\sqrt{t^2-5^2}}\,dt$$

$$= \frac{1}{2}\log\left|t + \sqrt{t^2-5^2}\right| + c$$

$$I = \frac{1}{2}\log\left|2x + \sqrt{4x^2-25}\right| + c$$

வகை II

$\int \frac{dx}{ax^2+bx+c}$ மற்றும் $\int \frac{dx}{\sqrt{ax^2+bx+c}}$ **எனும் வடிவில் உள்ள தொகைகளைக் காணல் :**